PICARD GROUP OF A TANGENT-VECTOR SQUISHING: EXACT SEQUENCE WITH ADDITIVE KERNEL CONTROLLED BY THE SQUISHED VECTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the surjectivity input, the units sequence with the Leray inclusion, the kernel identification, and the injectivity argument are all retained verbatim.

Editorial note. The label lemma-squish-tangent-vector names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Surjectivity for birational finite morphisms. Varieties, Lemma varieties-lemma-surjective-pic-birational-finite. Gives surjectivity of Pic(X) -> Pic(X') for the squishing morphism.
- The units exact sequence of a squishing. Same chapter, Example example-squish-tangent-vector; Cohomology, Lemma cohomology-lemma-Leray. Supplies 0 -> O_X^* -> f_*O_{X'}^* -> x_*k -> 0 with the theta map, and the Leray bound H^1(X, f_*O_{X'}^*) subset H^1(X', O_{X'}^*).
- Regular functions on proper reduced schemes. Varieties, Lemma varieties-lemma-proper-geometrically-reduced-global-sections. Identifies global regular functions with k over the algebraically closed field, giving the injectivity statement.

Source reference. Original lemma 01F2: sources/stacks-chow/tags.txt, source lines 1855–1855. Prerequisite pointer invoked at proof line(s) 3390; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1K: sources/stacks-curves/pinned-original-source.tex, source lines 3228–3228. Prerequisite pointer invoked at proof line(s) 3367; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BUG: sources/stacks-chow/tags.txt, source lines 14874–14874. Prerequisite pointer invoked at proof line(s) 3396; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0C0T: sources/stacks-chow/tags.txt, source lines 15098–15098. Prerequisite pointer invoked at proof line(s) 3377; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be an algebraically closed field. If $f : X' \to X$ is the squishing of a tangent vector ϑ as in Example 0C1K, then there is an exact sequence*

$$(k, +) \to \mathrm{Pic}(X) \to \mathrm{Pic}(X') \to 0$$

and the first map is injective if X' is proper and reduced.

Proof. The map $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$ is surjective by Varieties, Lemma 0C0T. Using the short exact sequence

$$0 \to \mathcal{O}_X^* \to f_*\mathcal{O}_{X'}^* \xrightarrow{\vartheta} x_*k \to 0$$

of Example 0C1K we obtain

$$H^0(X', \mathcal{O}_{X'}^*) \xrightarrow{\vartheta} k \to H^1(X, \mathcal{O}_X^*) \to H^1(X, f_*\mathcal{O}_{X'}^*)$$

We have $H^1(X, f_*\mathcal{O}_{X'}^*) \subset H^1(X', \mathcal{O}_{X'}^*)$ (for example by the Leray spectral sequence, see Cohomology, Lemma 01F2). Hence the kernel of $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$ is the cokernel of the map $\vartheta : H^0(X', \mathcal{O}_{X'}^*) \to k$. Because k is algebraically closed any regular function on a reduced connected proper scheme over k comes from an element of k, see Varieties, Lemma 0BUG. Thus the final statement of the lemma. □